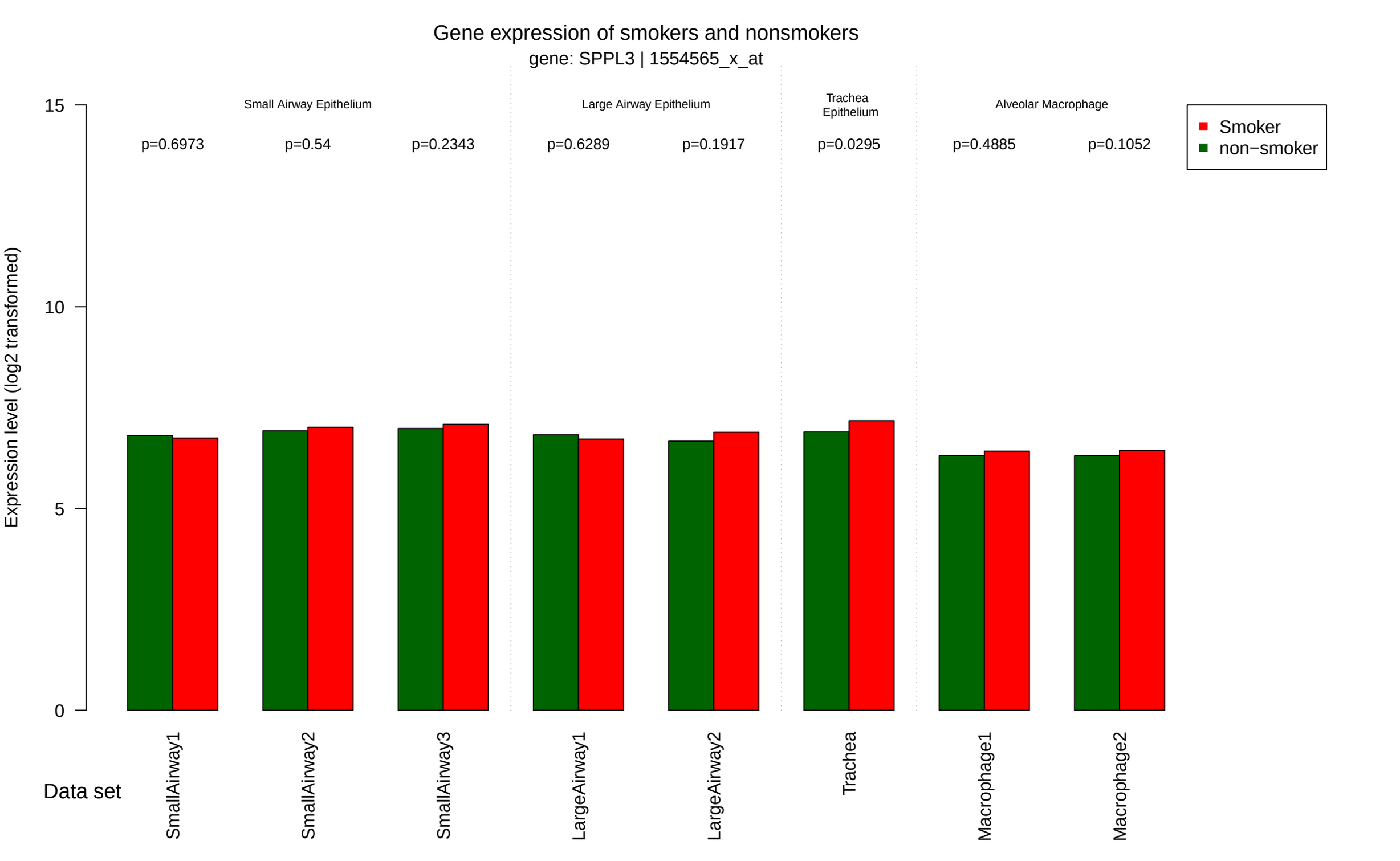
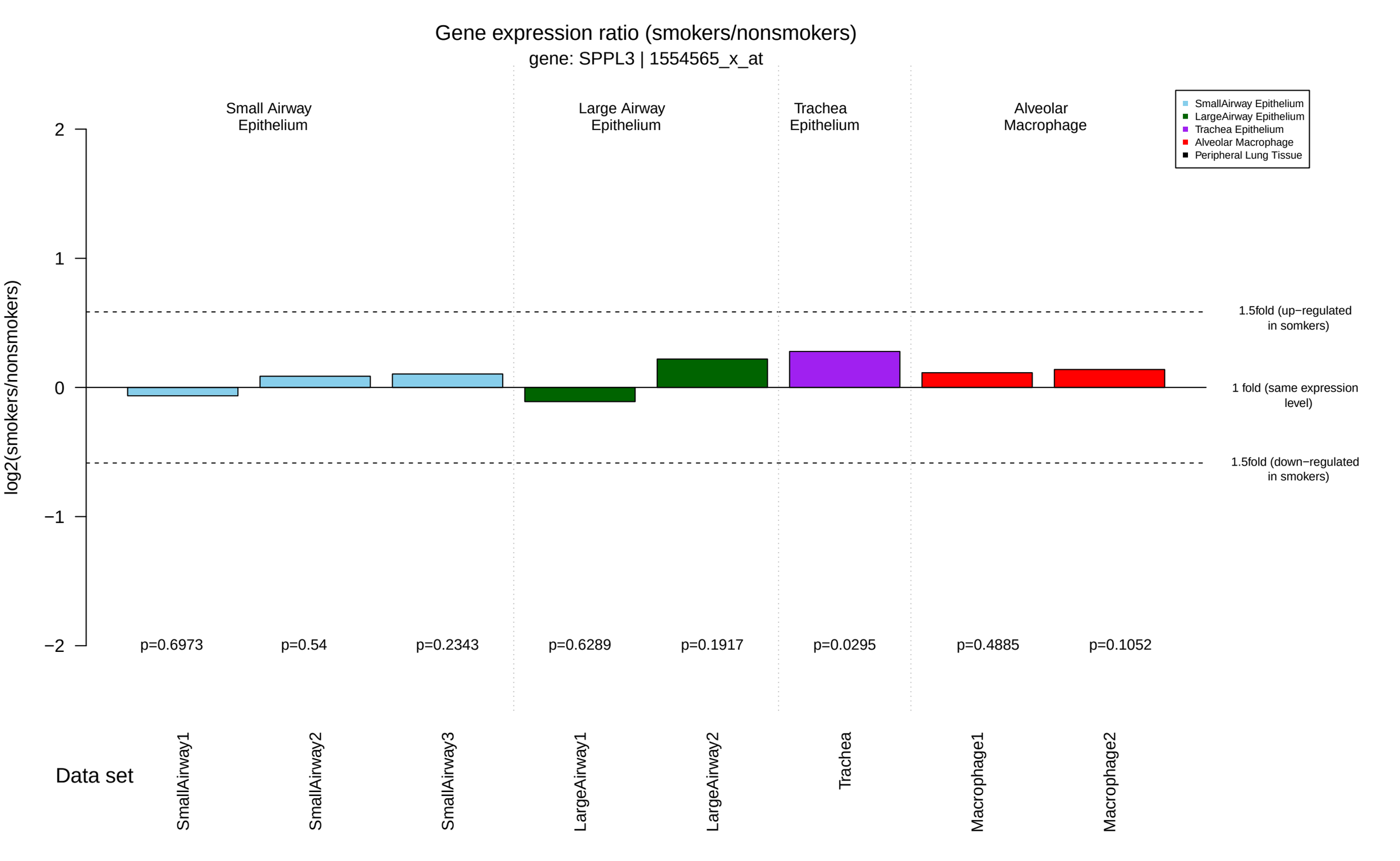
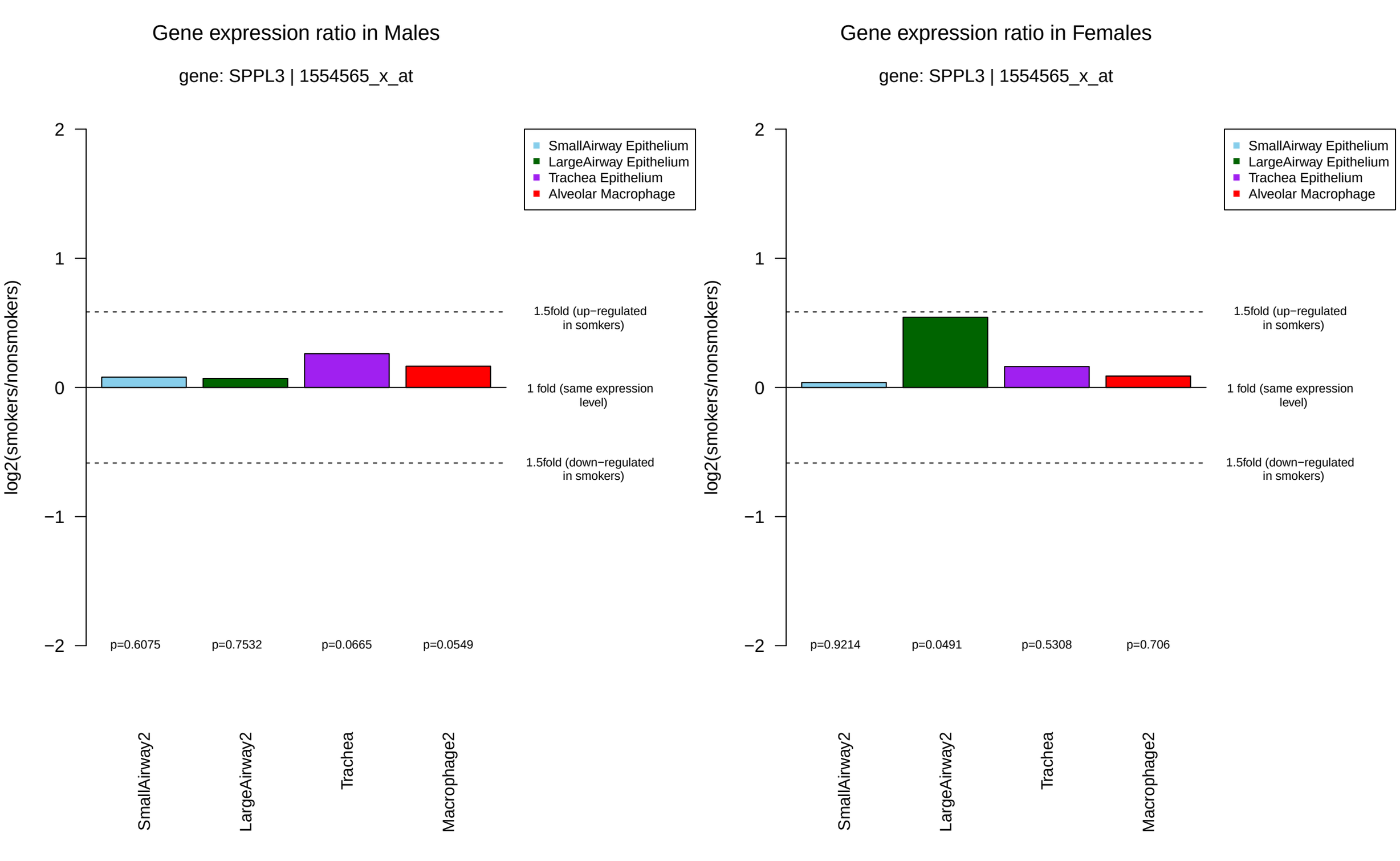
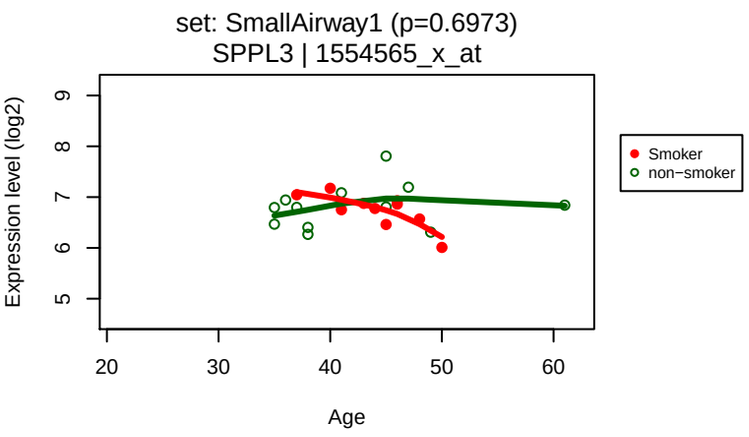
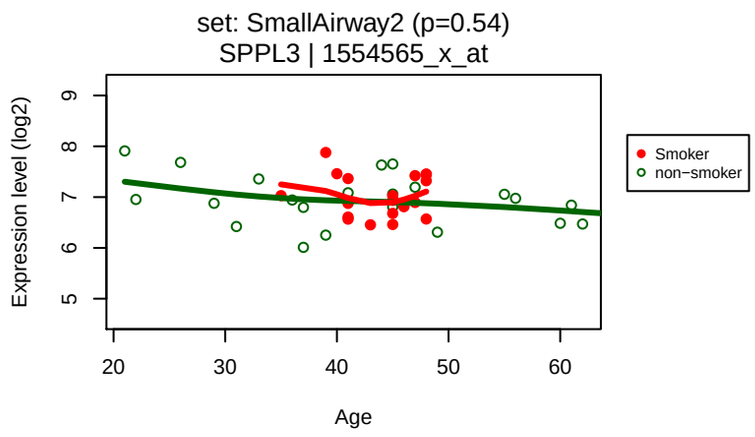
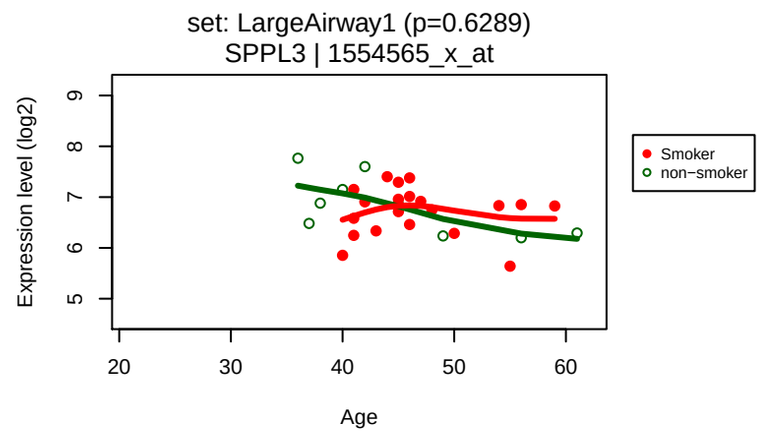

Gene expression of smokers and nonsmokers
gene: SPPL3 | 1554565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6973
p=0.54
p=0.2343
p=0.6289
p=0.1917
p=0.0295
p=0.4885
p=0.1052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

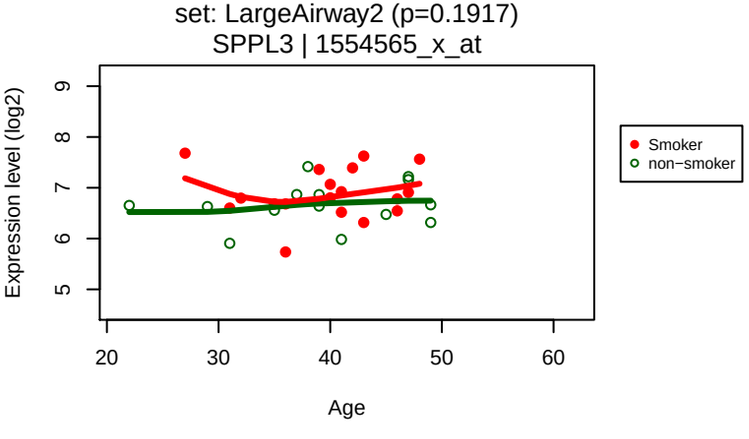

Gene expression ratio (smokers/nonsmokers)
gene: SPPL3 | 1554565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6973
p=0.54
p=0.2343
p=0.6289
p=0.1917
p=0.0295
p=0.4885
p=0.1052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

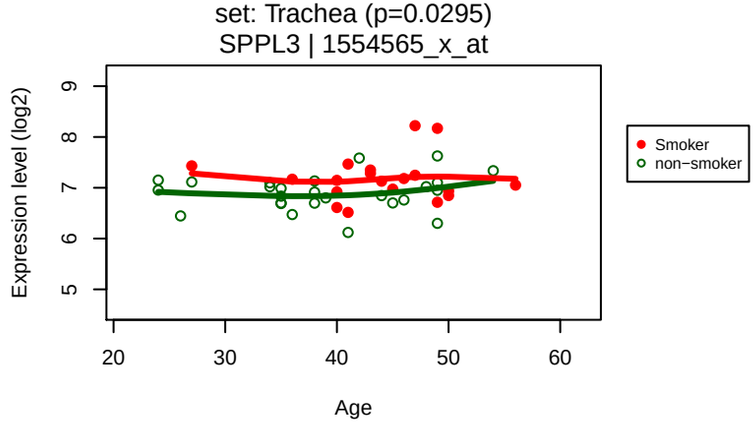

Gene expression ratio in Males
gene: SPPL3 | 1554565_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6075
p=0.7532
p=0.0665
p=0.0549
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPPL3 | 1554565_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9214
p=0.0491
p=0.5308
p=0.706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

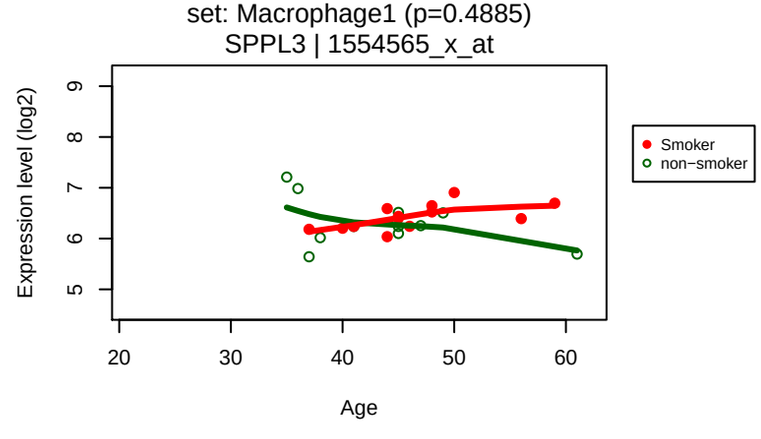

set: SmallAirway1 (p=0.6973)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker

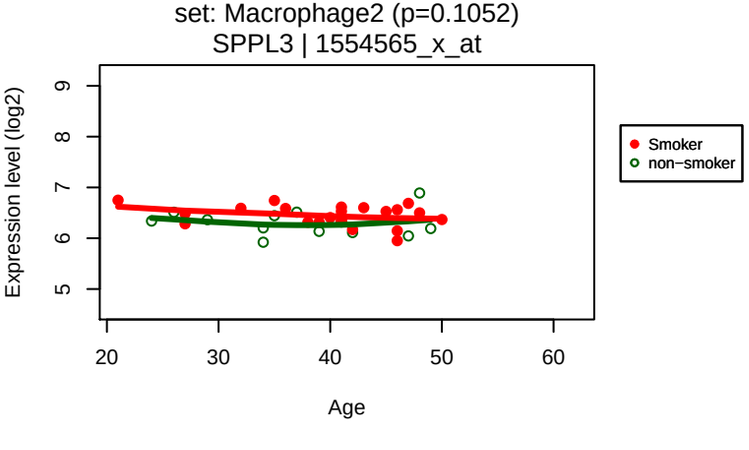

set: SmallAirway2 (p=0.54)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker

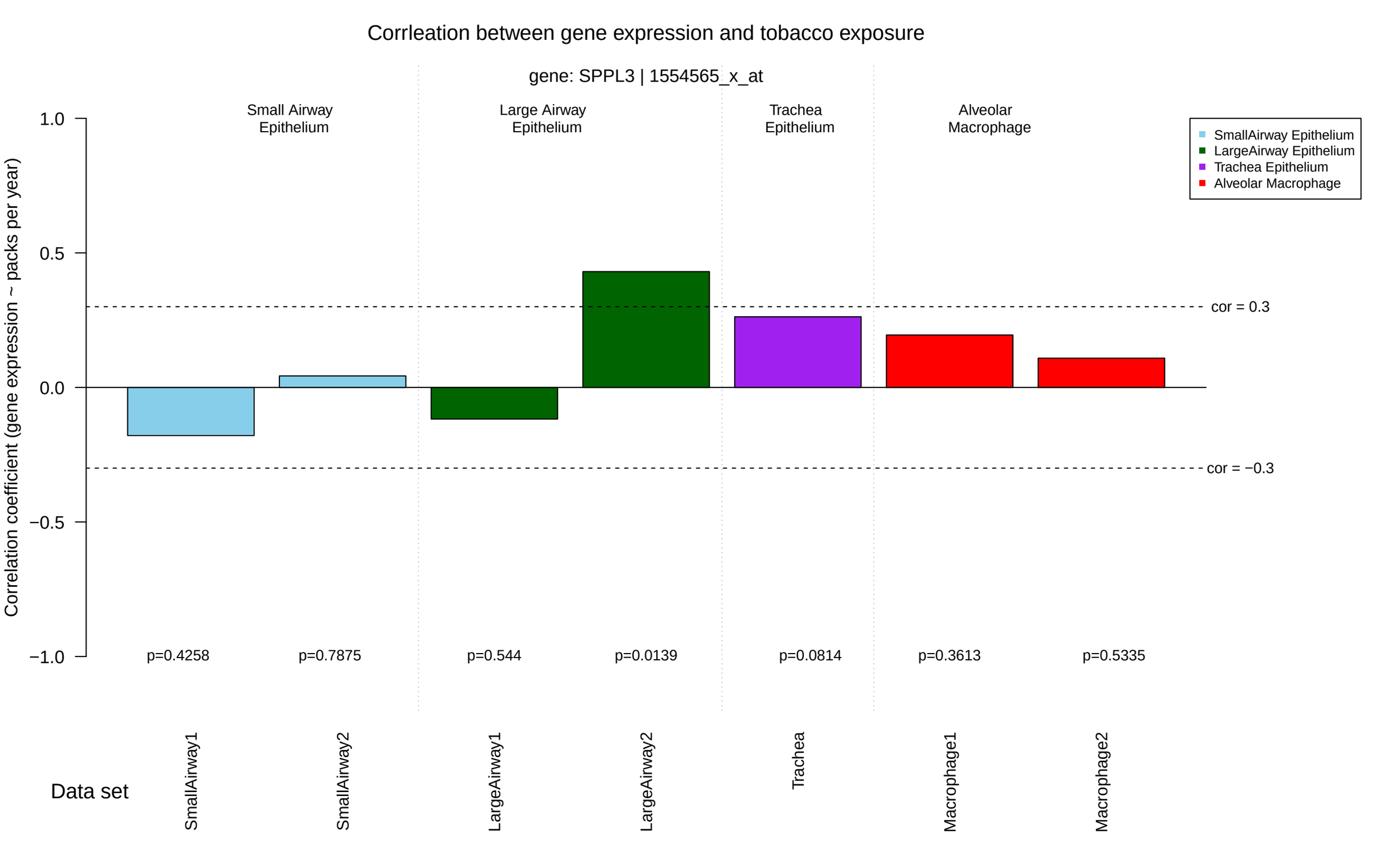

set: LargeAirway1 (p=0.6289)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1917)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0295)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4885)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1052)
SPPL3 | 1554565_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPPL3 | 1554565_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4258
p=0.7875
p=0.544
p=0.0139
p=0.0814
p=0.3613
p=0.5335
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2